M. M. ENGINEERING COLLEGE – MULLANA			LABORATORY FILE
EXPERIMENT TITLE: Configuration of Static Routing for Interconnection of Networks using Cisco Packet Tracer			
EXPERIMENT NO. BCSE-509L-06			**Execution Date** : 09/03/2026
SUBJECT : Computer Networks Lab		**SEMESTER** : IV	**Session:-** 2025 - 2026
Faculty:- Professor Munishwar Rai			

Aim

To study and configure static routing for interconnection and communication between different networks using Cisco Packet Tracer.

Requirements

- Cisco Packet Tracer
- Routers
- Switches
- PCs
- Connecting cables

Theory

Static routing is a method in which routes are manually configured in the router. It is used to forward data packets from one network to another. In this method, the network administrator defines the path for data transmission.

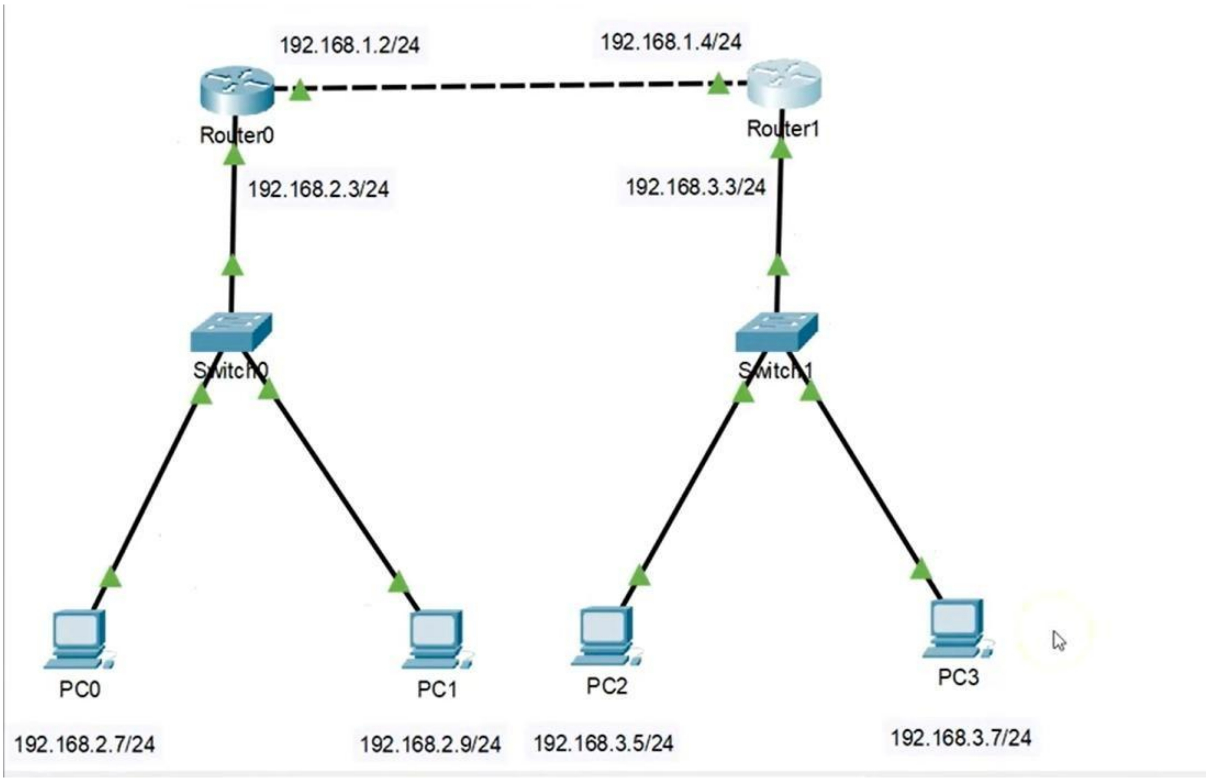
Routers are used to interconnect different networks, and static routes are configured to enable communication between them.

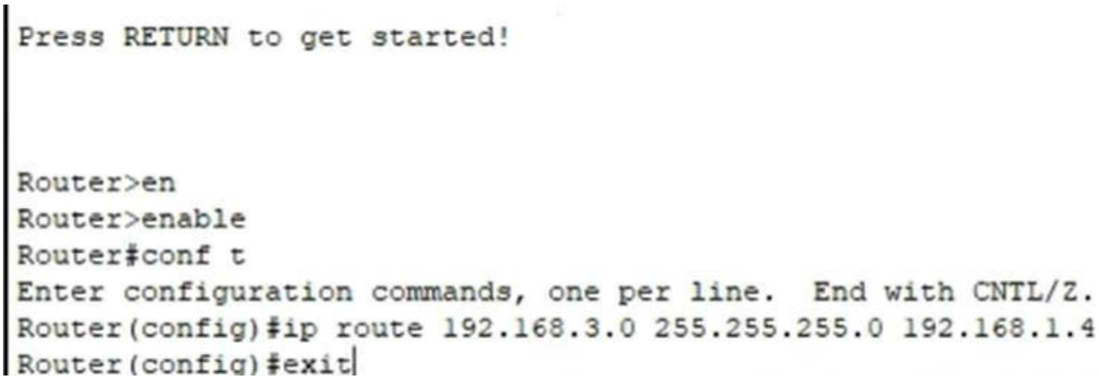
Procedure

1. Open Cisco Packet Tracer.
2. Place routers, switches, and PCs in the workspace.
3. Connect devices using appropriate cables.
4. Assign IP addresses to all PCs and router interfaces.
5. Configure router interfaces using CLI.
6. Apply static routing commands on routers.

7. Save configuration.

8. Test connectivity using ping command.





```
Press RETURN to get started!



Router>en
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.3.0 255.255.255.0 192.168.1.4
Router(config)#exit
```

Router0 side

- Network: **192.168.2.0/24**
- Router IP: **192.168.2.3**
- PCs:
  - PC0 → 192.168.2.7
  - PC1 → 192.168.2.9

```
Router>en
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.1.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
Router#
```

Router1 side

- Network: **192.168.3.0/24**
- Router IP: **192.168.3.3**
- PCs:
  - PC2 → 192.168.3.5
  - PC3 → 192.168.3.7

Router to Router link

- Network: **192.168.1.0/24**
- Router0 → 192.168.1.2
- Router1 → 192.168.1.4

Viva Questions – Static Routing

1. What is static routing?

Static routing is a method where routes are manually configured in a router.

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2. Which software did you use?

Cisco Packet Tracer.

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3. Why is static routing used?

It is used to define a fixed path for data between networks.

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4. What is the syntax of static routing?

Ip route destination-network subnet-mask next-hop

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5. What is a next hop?

It is the IP address of the next router where the packet is forwarded.

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6. How many networks are in your diagram?

There are 3 networks:

- 192.168.2.0
- 192.168.3.0
- 192.168.1.0

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7. What is the advantage of static routing?

It is simple, secure, and uses less bandwidth.

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8. What is the disadvantage of static routing?

It is not scalable and needs manual configuration.

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9. What is the role of a router?

A router connects different networks and forwards data packets.

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10. How do you test connectivity?

Using the ping command.

EXPERIMENT TITLE: Configuration of Distance Vector RIP Routing Protocol using Cisco Packet Tracer.		
EXPERIMENT NO. BCSE-509L-07		**Execution Date** : 16/03/2026
SUBJECT : Computer Networks Lab	**SEMESTER** : IV	**Session:-** 2025 - 2026
Faculty:- Professor Munishwar Rai		

Aim

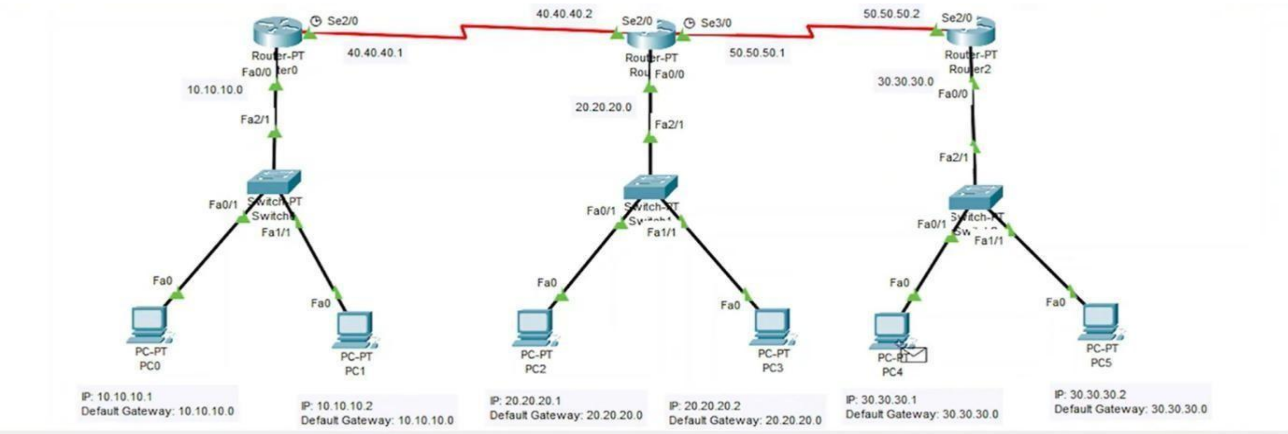
To study and configure the Distance Vector Routing Protocol (RIP) for communication between different networks using Cisco Packet Tracer.

Requirements

- Cisco Packet Tracer
- Routers
- Switches
- PCs
- Connecting cables

Procedure

1. Open Cisco Packet Tracer.
2. Place routers, switches, and PCs.
3. Connect all devices properly.
4. Assign IP addresses to PCs and router interfaces.
5. Configure router interfaces using CLI.
6. Enable RIP routing on routers.
7. Add network addresses in RIP configuration.
8. Save configuration.
9. Test connectivity using ping command.

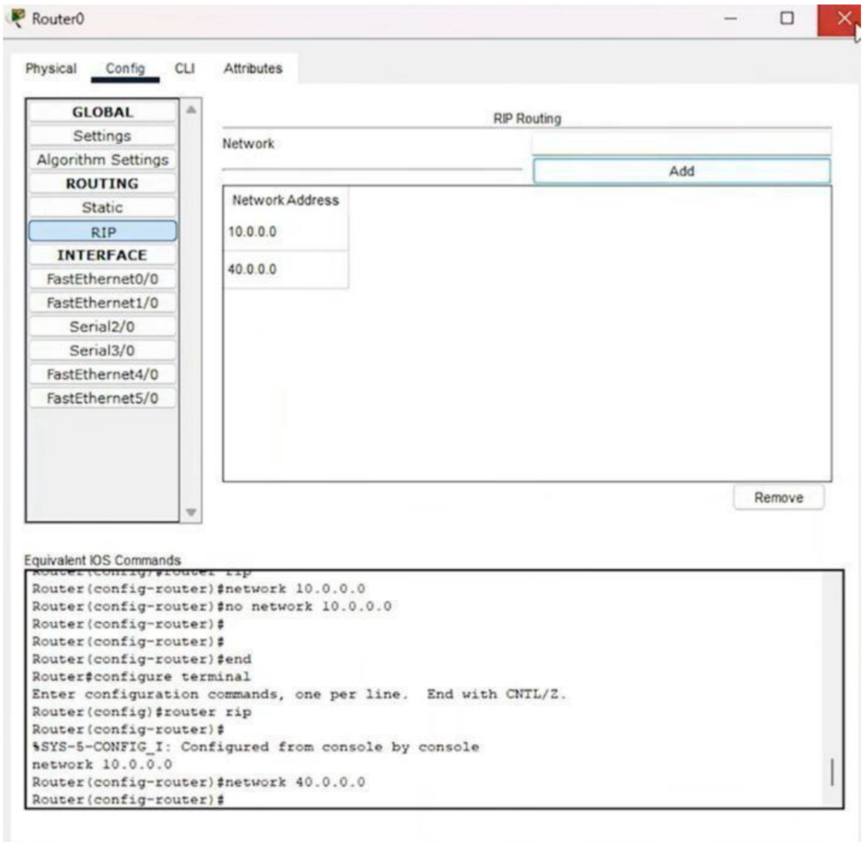


Device	Interface	IP Address Network
Router0 Fa0/0	10.10.10.0	10.0.0.0
Router0 Se2/0	40.40.40.1	40.0.0.0
Router1 Fa0/0	20.20.20.0	20.0.0.0
Router1 Se2/0	40.40.40.2	40.0.0.0
Router1 Se3/0	50.50.50.1	50.0.0.0
Router2 Fa0/0	30.30.30.0	30.0.0.0
Router2 Se2/0	50.50.50.2	50.0.0.0

RIP Configuration Commands

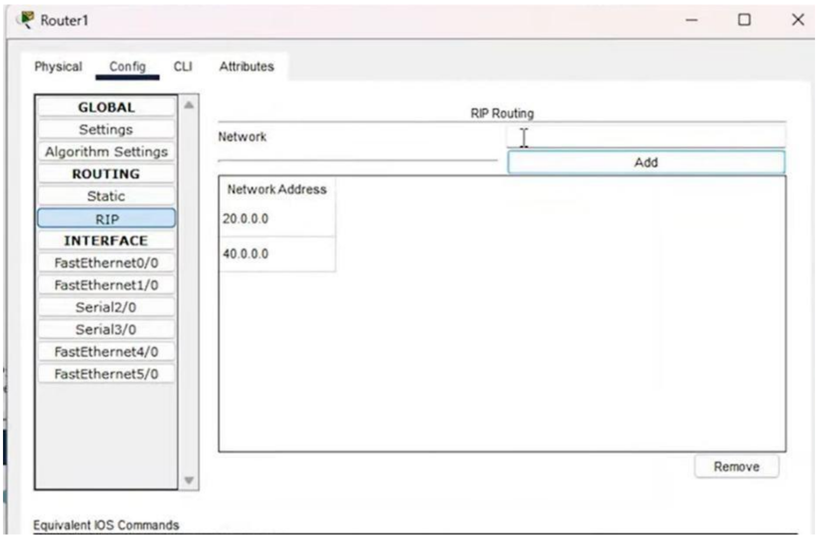
Router0:

- Router> enable
- Router# configure terminal
- Router(config)# router rip
- Router(config-router)# network 10.0.0.0
- Router(config-router)# network 40.0.0.0
- Router(config-router)# end



Router1

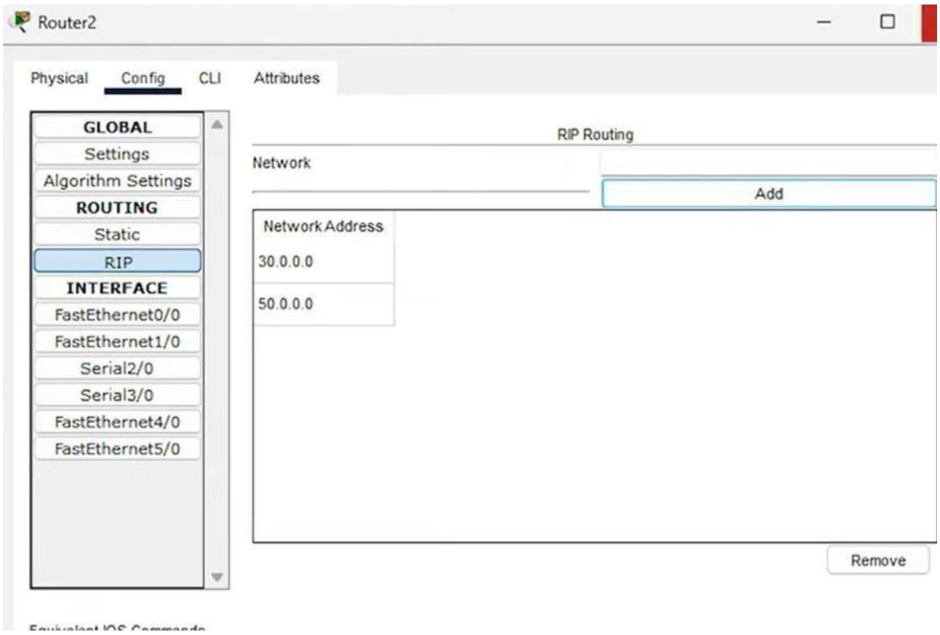
- Router> enable
- Router# configure terminal
- Router(config)# router rip
- Router(config-router)# network 20.0.0.0 • Router(config-router)# network 40.0.0.0
- Router(config-router)# network 50.0.0.0
- Router(config-router)# end



Router2:

- Router> enable
- Router# configure terminal
- Router(config)# router rip

- Router(config-router)# network 30.0.0.0
- Router(config-router)# network 50.0.0.0
- Router(config-router)# end



Viva Questions – RIP Routing Protocol

1. What is RIP?

RIP (Routing Information Protocol) is a distance vector routing protocol used to find the best path between networks.

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2. Which software did you use?

Cisco Packet Tracer.

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3. What is the metric used in RIP?

RIP uses **hop count** as a metric.

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4. What is the maximum hop count in RIP?

The maximum hop count is **15**.

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5. What happens if hop count exceeds 15?

The network becomes unreachable.

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6. What type of routing protocol is RIP?

It is a **dynamic routing protocol**.

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7. What is the difference between RIP v1 and RIP v2?

RIP v1 is classful, while RIP v2 is classless and supports subnetting.

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8. How often does RIP send updates?

Every **30 seconds**.

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9. What command is used to enable RIP?

router rip

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10. How do you test network connectivity?

Using the **ping command**.